

Multivariable Calculus

Unit 1
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1 Vectors and the Geometry of Space

1.1 Three-Dimensional Coordinate Systems

- A **plane** is two-dimensional because any point (x, y) on the plane can be represented using only two real numbers.
- A **space** is a three-dimensional collection of points, where each point is represented using three real numbers (x, y, z) .
 - We can set three orthogonal (perpendicular) lines that pass through a given origin $\mathcal{O} = (0, 0, 0)$. We can then call each of these lines the x -, y -, and z -axes respectively.
 - Typically, we imagine that the xy -plane (the plane which lies on the x - and y -axes) is at the bottom:
 - the x - and y -axes are “horizontal”.
 - the z -axis is “vertical”, with “upwards” indicating the positive z -direction.
 - The direction of the z -axis can be determined using the **right-hand rule**:
Extend your right hand in the direction of $+x$, then curl your fingers in the direction of $+y$. Your thumb will point in the direction of $+z$.
 - There are three **coordinate planes**: the xy -, xz -, and yz -planes.
 - The **projection** of a point P to a specific plane is the coordinates of P with the “other” coordinate set to 0.
ex: the projection of $(2, 5, 3)$ to the xz -axis is $(2, 0, 3)$. The y -coordinate is set to 0.
- n -**space** is the n -dimensional space represented by the collection of points represented by a collection of n real numbers.
- $\mathbb{R}^n$ is the set of all points in n -space.
 $\mathbb{R}^2$ is the set of all points in a plane, and $\mathbb{R}^3$ is the set of all points in 3-space.
 - If there is a bijection from coordinates to points in space (if a point in space can only ever be represented with one coordinate and vice versa), the coordinate system is **rectangular**.

Graphs in 3-space

- In 2-space, the set of points that satisfy an equation (e.g. one involving x and y) is called a **curve** in $\mathbb{R}^2$.
- Similarly, the set of points that satisfy equation in 3-space (e.g. one involving x , y , and z) is called a **surface** in $\mathbb{R}^3$.
- The following equations in $\mathbb{R}^3$ represent planes parallel to one of the coordinate planes:

$$x = c \quad y = c \quad z = c \quad \text{where } c \text{ is a constant.}$$

For example, the equation $z = 3$ represents all points $(x, y, 3)$, which is the xy -plane translated 3 units up (in the positive z -direction).

Distance between Two Points

The distance between two points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ in 3-space is:

$$|PQ| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

Using the distance formula, we can derive the equation of the sphere of radius r to be the set of all points $P(x, y, z)$ a distance r from some center $\mathcal{C}(h, k, \ell)$:

$$\begin{aligned} |P\mathcal{C}| &= r \\ \sqrt{(x - h)^2 + (y - k)^2 + (z - \ell)^2} &= r \\ (x - h)^2 + (y - k)^2 + (z - \ell)^2 &= r^2. \end{aligned}$$

Example: Finding the center and radius of a sphere

Find the center and radius of the sphere $x^2 + y^2 + z^2 + 4x - 6y + 2z + 6 = 0$.

Rewrite into standard form by completing the squares:

$$\begin{aligned} (x^2 + 4x) + (y^2 - 6y) + (z^2 + 2z) + 6 &= 0 \\ (x^2 + 4x + 4) + (y^2 - 6y + 9) + (z^2 + 2z + 1) + 6 &= 4 + 9 + 1 \\ (x + 2)^2 + (y - 3)^2 + (z + 1)^2 &= 4 + 9 + 1 - 6 \\ (x + 2)^2 + (y - 3)^2 + (z + 1)^2 &= 8 = \underbrace{\sqrt{8}^2}_r \end{aligned}$$

Thus, the center is $(-2, 3, -1)$ and the sphere has radius $r = \sqrt{8} = 2\sqrt{2}$.

1.2 Introduction to Vectors

- Vectors are either denoted with arrows or they are bolded.
 - If v is a vector, then we can clarify so by writing $\vec{v}$ or $\mathbf{v}$.
- If the points A and B are defined, the vector pointing from A to B can be written $\overrightarrow{AB}$.
- A vector in n -dimensions is composed of n *components*. Each vector component represents a certain direction in space.
- A vector can be written in component form, either as a 1-D matrix (a “column vector”) or by using angle brackets:

$$\vec{v} = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix} = \langle v_1, v_2, \dots, v_n \rangle.$$

- The **magnitude** of a vector is the distance from head to tail. The magnitude of a vector $\vec{v}$ is denoted $|\vec{v}|$ or $\|\vec{v}\|$. If $\vec{v} = \langle v_1, v_2, \dots, v_n \rangle$, then its magnitude $|\vec{v}| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$.
- A **unit vector** is a vector with magnitude 1. If $\vec{v}$ is a well-defined unit vector, then it is typically denoted $\hat{v}$ (“*v-hat*”), and $|\hat{v}| = 1$.
 - The three *basis vectors*, or fundamental directional unit vectors, are $\hat{i}$, $\hat{j}$, and $\hat{k}$. It holds that $\langle x, y, z \rangle = x\hat{i} + y\hat{j} + z\hat{k}$.
 - The unit vector $\hat{v}$ of a vector $\vec{v}$ points in the same direction as $\vec{v}$, and $\hat{v} = \frac{\vec{v}}{|\vec{v}|}$.
- The **zero vector** is the only vector with a magnitude of 0, denoted $\vec{0}$.
- A vector only describes a displacement, not an absolute position. A vector has no way of describing where it starts and ends without some reference point.

Combining Vectors

- The sum of two vectors $\vec{a} = \langle a_1, a_2, \dots, a_n \rangle$ and $\vec{b} = \langle b_1, b_2, \dots, b_n \rangle$ is:

$$\vec{a} + \vec{b} = \langle a_1, a_2, \dots, a_n \rangle + \langle b_1, b_2, \dots, b_n \rangle = \langle a_1 + b_1, a_2 + b_2, \dots, a_n + b_n \rangle.$$

In other words, it is just adding each corresponding component together.

- The difference of two vectors works the same way.
- It holds that $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$ (triangle law) and $\vec{a} + \vec{b} = \vec{b} + \vec{a}$ (parallelogram law), each named for how the property can be visualized on a graph.

Scalar Products

Multiplying a vector by a scalar (or vice-versa) is the equivalent of multiplying each component by the scalar:

$$k\vec{v} = k\langle v_1, v_2, \dots, v_n \rangle = \langle kv_1, kv_2, \dots, kv_n \rangle \quad \text{where } k \text{ is any scalar.}$$

It also holds that $|k\vec{v}| = k|\vec{v}|$.

1.3 Products of Vectors

There are two fundamental ways to “combine” two vectors: by using a **dot product** and by using a **cross product**.

The Dot Product

The dot product of two vectors $\vec{a}$ and $\vec{b}$, denoted $\vec{a} \cdot \vec{b}$, interprets the product as both how much the two vectors point in the same direction as well as the scalar product of their magnitudes.

- Geometrically, if θ is the angle between the vectors $\vec{a}$ and $\vec{b}$, then $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos \theta$.
 - Thus, if $\vec{a}$ is orthogonal to $\vec{b}$, then $\vec{a} \cdot \vec{b} = 0$ because $\cos 90^\circ = 0$.
 - Additionally, if $\vec{a}$ points in the same direction as $\vec{b}$, then $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}|$ because $\cos 0^\circ = 1$.
- For example, work can be defined as the dot product of force $\mathbf{F}$ and displacement $\mathbf{r}$, since only the component of force directed along the displacement should be considered ($W = \mathbf{F} \cdot \mathbf{r}$).

Derivation: Dot Product using Components

The first principles of the dot product is actually the geometric definition of the dot product:

$$\vec{a} \cdot \vec{b} = ab \cos \theta$$

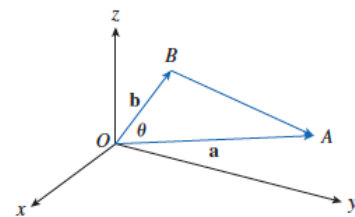
...where $a = |\vec{a}|$, $b = |\vec{b}|$, and θ is the angle between $\vec{a}$ and $\vec{b}$ ($0 \leq \theta \leq \pi$).

It turns out that a , b , and $|\vec{a} - \vec{b}|$ are vectors that make up the side lengths of a triangle (see right figure).

Let $\vec{a} = \langle a_1, a_2, \dots, a_n \rangle$ and $\vec{b} = \langle b_1, b_2, \dots, b_n \rangle$.

By the law of cosines:

$$\begin{aligned} |\vec{a} - \vec{b}|^2 &= a^2 + b^2 - 2ab \cos \theta \\ &= a^2 + b^2 - 2(\vec{a} \cdot \vec{b}). \end{aligned}$$



We can then isolate the dot product:

$$\begin{aligned} \vec{a} \cdot \vec{b} &= \frac{1}{2} \left(a^2 + b^2 - |\vec{a} - \vec{b}|^2 \right) \\ &= \frac{1}{2} \left(a_1^2 + a_2^2 + \dots + a_n^2 \right. \\ &\quad \left. + b_1^2 + b_2^2 + \dots + b_n^2 \right. \\ &\quad \left. - (a_1 - b_1)^2 - (a_2 - b_2)^2 - \dots - (a_n - b_n)^2 \right) \\ &= \frac{1}{2} \left(a_1^2 + a_2^2 + \dots + a_n^2 \right. \\ &\quad \left. + b_1^2 + b_2^2 + \dots + b_n^2 \right. \\ &\quad \left. - (a_1^2 - 2a_1b_1 + b_1^2) - (a_2^2 - 2a_2b_2 + b_2^2) - \dots - (a_n^2 - 2a_nb_n + b_n^2) \right) \\ &= \frac{1}{2} (2a_1b_1 + 2a_2b_2 + \dots + 2a_nb_n) = a_1b_1 + a_2b_2 + \dots + a_nb_n. \end{aligned}$$

Dot Product of Two Vectors (Component Form)

Let $\vec{a} = \langle a_1, a_2, \dots, a_n \rangle$ and $\vec{b} = \langle b_1, b_2, \dots, b_n \rangle$ be vectors in n -dimensions. Then:

$$\vec{a} \cdot \vec{b} = \sum_{i=1}^n a_i b_i = a_1 b_1 + a_2 b_2 + \dots + a_n b_n.$$

The Cross Product

Cross Product of Two Vectors in 3D

Let $\vec{a} = \langle a_x, a_y, a_z \rangle$ and $\vec{b} = \langle b_x, b_y, b_z \rangle$. Then:

$$\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix} = \begin{pmatrix} a_y b_z - a_z b_y \\ a_z b_x - a_x b_z \\ a_x b_y - a_y b_x \end{pmatrix}$$

- Unlike dot products which output scalar values, cross products output vectors.
 - The *right hand rule* can be used to determine the direction of the resulting vector.
If the vectors $\vec{a}$ and $\vec{b}$ lie on a plane P , then $\vec{a} \times \vec{b}$ is orthogonal to P .
 - The unit vector pointing in the direction of the resulting vector is typically denoted $\hat{n}$. This is called the *normal vector* or the *unit normal*.
- Geometrically, if θ is the angle between the vectors $\vec{a}$ and $\vec{b}$, then $\vec{a} \times \vec{b} = (|\vec{a}||\vec{b}|\sin\theta)\hat{n}$.
 - Thus, if $\vec{a}$ points in the same direction as $\vec{b}$, then $\vec{a} \times \vec{b} = 0$ because $\sin 0^\circ = 0$.
This makes sense because $\vec{a}$ and $\vec{b}$ no longer form a plane (they are not *coplanar*).
 - Additionally, if $\vec{a}$ is orthogonal to $\vec{b}$, then $|\vec{a} \times \vec{b}| = |\vec{a}||\vec{b}|$ because $\sin 90^\circ = 1$.
- **Cross products are not commutative:** $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$
(if we flip the order, the cross product will point in the opposite direction)
- **Cross products are not associative:** $(\vec{a} \times \vec{b}) \times \vec{c} \neq \vec{a} \times (\vec{b} \times \vec{c})$.
- Cross products are an interpretation of the area of the parallelogram formed by the vectors.
 $|\vec{a} \times \vec{b}|$ is the area of the parallelogram bounded by $\vec{a}$ and $\vec{b}$!

Example: Area of a Triangle given Three Points

Find the area of the triangle ABC with vertices $A(2, 1, 2)$, $B(3, 5, 4)$, and $C(1, 6, 7)$.

Although we can try computing the lengths of each side of the triangle and apply triangle formulas to solve this problem, we can just as well use a cross product.

The area of the parallelogram with adjacent sides AB and BC is $|\overrightarrow{AB} \times \overrightarrow{BC}|$.

Find $\overrightarrow{AB} \times \overrightarrow{BC}$ and compute its magnitude:

$$\begin{aligned}\overrightarrow{AB} \times \overrightarrow{BC} &= \langle 3 - 2, 5 - 1, 4 - 2 \rangle \times \langle 1 - 3, 6 - 5, 7 - 4 \rangle \\ &= \langle 1, 4, 2 \rangle \times \langle -2, 1, 3 \rangle \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 4 & 2 \\ -2 & 1 & 3 \end{vmatrix} = \langle 10, -7, 9 \rangle.\end{aligned}$$

$$\text{Area}_{\text{parallelogram}} = |\overrightarrow{AB} \times \overrightarrow{BC}| = \sqrt{10^2 + (-7)^2 + 9^2} = \sqrt{230}.$$

The area of the triangle is half the area of the parallelogram:

$$\text{Area}_{\triangle ABC} = \frac{1}{2} \text{Area}_{\text{parallelogram}} = \frac{\sqrt{230}}{2}.$$

Cross Products of Basis Vectors

You should use the right-hand rule to memorize the following cross products:

$$\begin{array}{ll}\hat{i} \times \hat{j} = \hat{k} & \hat{j} \times \hat{i} = -\hat{k} \\ \hat{j} \times \hat{k} = \hat{i} & \hat{k} \times \hat{j} = -\hat{i} \\ \hat{k} \times \hat{i} = \hat{j} & \hat{i} \times \hat{k} = -\hat{j}\end{array}$$

Triple Products

- The **scalar triple product** of three vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ is $\vec{a} \cdot (\vec{b} \times \vec{c})$. The triple product is relevant since it is the volume of the parallelepiped (parallelogram prism) formed by the three vectors.

If the volume of the parallelepiped is 0, then the triple product is 0 and our vectors lie on the same plane (the parallelepiped is flat). In other words, $\vec{a}$, $\vec{b}$, and $\vec{c}$ are **coplanar** if $\vec{a} \cdot (\vec{b} \times \vec{c}) = 0$.

Conveniently, it also holds that $\vec{a} \cdot (\vec{b} \times \vec{c}) = (\vec{a} \times \vec{b}) \cdot \vec{c} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$.

- The **vector triple product** of three vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ is $\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$.

1.4 Projections of Vectors

The **projection** of a vector $\vec{a}$ along another vector $\vec{b}$ is the component of $\vec{a}$ which points in the same direction as $\vec{b}$. There are two types of projections:

- The **scalar projection** is simply the magnitude of such component. We can derive it from the definition of the dot product:

Derivation: Scalar Projection

1. Let $a = |\vec{a}|$ and $b = |\vec{b}|$.
2. Since $\vec{a} \cdot \vec{b} = ab \cos \theta$, where θ is the angle between $\vec{a}$ and $\vec{b}$, $\cos \theta = \frac{\vec{a} \cdot \vec{b}}{ab}$.
3. Using trigonometry, $\cos \theta = \frac{x}{a}$, where x is the scalar component of $\vec{a}$ along $\vec{b}$.
4. $\therefore \frac{x}{a} = \frac{\vec{a} \cdot \vec{b}}{ab} \implies x = \frac{\vec{a} \cdot \vec{b}}{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$.

- The scalar projection is often represented using the syntax $\text{comp}_{\vec{b}} \vec{a}$, where $\vec{b}$, the subscript, is what we are projecting *onto*.

Scalar Projection

The **scalar projection** of $\vec{a}$ along (“onto”) $\vec{b}$, $\text{comp}_{\vec{b}} \vec{a}$, is the signed magnitude of the component of $\vec{a}$ which is parallel to $\vec{b}$:

$$\text{comp}_{\vec{b}} \vec{a} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$$

- The **vector projection** of $\vec{a}$ along $\vec{b}$, $\text{proj}_{\vec{b}} \vec{a}$, is the vector component of $\vec{a}$ pointing in the same direction as $\vec{b}$. It is simply the scalar projection times the unit vector for $\vec{b}$:

$$\text{proj}_{\vec{b}} \vec{a} = \text{comp}_{\vec{b}} \vec{a} \cdot \hat{b} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|} \cdot \frac{\vec{b}}{|\vec{b}|} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \vec{b}.$$

Vector Projection

The **vector projection** of $\vec{a}$ along (“onto”) $\vec{b}$, $\text{proj}_{\vec{b}} \vec{a}$, is the vector component of $\vec{a}$ pointing in the same direction as $\vec{b}$:

$$\text{proj}_{\vec{b}} \vec{a} = \frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \vec{b}.$$

- **proj** is often bolded because the operation results in a vector.
- The result of the projection specified by **proj** is in the direction of the *subscript*.

1.5 Equations of Lines and Planes

Point-Slope Equations of Lines

We know that in the xy -plane, we can define a line using its *point-slope form*, which specifies a point through which the line passes as well as the slope of the line.

A line which passes through the point (x_0, y_0) with slope m can be described using the equation $y = m(x - x_0) + y_0$. This is derived from applying translations to $y = mx$.

In a similar way, we can define a line in 3-space using a point through which the line passes as well as a *vector* which lies parallel to the line.

(In a way, our line has “multiple” slopes, which are all contained in the vector.)

Derivation: Parametric Equations for a Line in 3-space

1. Let the line L in 3-space pass through the point (x_0, y_0, z_0) . We can define the point using a **position vector**, $\vec{r}_0 = \langle x_0, y_0, z_0 \rangle$.

Now, assume the same line L points in the same direction as some vector $\vec{v} = \langle a, b, c \rangle$. We call a , b , and c the **direction numbers** for L .

Finally, let (x, y, z) be an arbitrary point on L , and $\vec{r} = \langle x, y, z \rangle$ is its position vector.

2. We can arrive at $\vec{r}$ by stepping in the direction of $\vec{v}$ starting from $\vec{r}_0$:

$$\vec{r} = \vec{r}_0 + t\vec{v}.$$

We introduce the parameter $t \in \mathbb{R}$ as an arbitrary scale factor of $\vec{v}$, since $\vec{v}$ does not specify the correct magnitude, only direction. Writing out the vector components:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x_0 \\ y_0 \\ z_0 \end{pmatrix} + t \begin{pmatrix} a \\ b \\ c \end{pmatrix} = \begin{pmatrix} x_0 + at \\ y_0 + bt \\ z_0 + ct \end{pmatrix}$$

3. This is a set of **parametric equations** which define the line L :

$$x(t) = x_0 + at \quad y(t) = y_0 + bt \quad z(t) = z_0 + ct$$

Point-Slope Equation of a Line

Let L be the line which passes through the point (x_0, y_0, z_0) and lies parallel to some vector $\vec{v} = \langle a, b, c \rangle$. Then a point (x, y, z) lies on L iff:

$$x(t) = x_0 + at \quad y(t) = y_0 + bt \quad z(t) = z_0 + ct \quad \text{where } t \in \mathbb{R}.$$

If $\vec{r} = \langle x, y, z \rangle$ and $\vec{r}_0 = \langle x_0, y_0, z_0 \rangle$, then L can also be represented by this vector equation:

$$\vec{r} = \vec{r}_0 + t\vec{v}.$$

Note

There can be many different sets of parametric equations which define the same lines in 3-space. We can change the point or choose a completely different direction vector.

Example: Equations of a Line from Point and Direction

Write a set of parametric equations which define the line that passes through the point $(5, 1, 3)$ and points in the same direction as $\hat{i} + 4\hat{j} - 2\hat{k}$.

Write each vector in component form:

$$\begin{aligned}\vec{r} &= \langle x, y, z \rangle \\ \vec{r}_0 &= \langle x_0, y_0, z_0 \rangle = \langle 5, 1, 3 \rangle \\ \vec{v} &= \langle a, b, c \rangle = \langle 1, 4, -2 \rangle\end{aligned}$$

From $\vec{r} = \vec{r}_0 + t\vec{v}$, we get the following parametric equations:

$$x(t) = 5 + t \qquad y(t) = 1 + 4t \qquad z(t) = 3 - 2t$$

Example: Equations of a Line from two Points

Write a set of parametric equations which define the line L that passes through the points $(6, 2, 7)$ and $(1, 4, 5)$. At what point does L intersect the xy -plane?

This is a simple extension from the content above, since the direction vector of a line passing through two points represented by position vectors $\vec{a}$ and $\vec{b}$ is just $\vec{a} - \vec{b}$ or $\vec{b} - \vec{a}$.

Let's choose $(1, 4, 5)$ as our given point, $\vec{r}_0 = \langle 1, 4, 5 \rangle$.

The direction vector can be computed as $\vec{v} = \langle 6, 2, 7 \rangle - \langle 1, 4, 5 \rangle = \langle 5, -2, 2 \rangle$.

We now have everything to write the parametric equations:

$$x(t) = 1 + 5t \qquad y(t) = 4 - 2t \qquad z(t) = 5 + 2t$$

The line will intersect the xy -plane when $z = 0$:

$$z(t) = 5 + 2t = 0 \implies t = -\frac{5}{2}.$$

Plug in $t = -\frac{5}{2}$ to every other equation in our parametric to find the point of intersection:

$$\begin{aligned}x\left(-\frac{5}{2}\right) &= 1 + 5\left(-\frac{5}{2}\right) = -\frac{23}{2} \\ y\left(-\frac{5}{2}\right) &= 4 - 2\left(-\frac{5}{2}\right) = 9\end{aligned}$$

(we already know $z = 0$)

Thus, the point that lies on L and intersects the xy -plane is $\left(-\frac{23}{2}, 9, 0\right)$.

Symmetric Equations of Lines

For each parametric equation, we can isolate t :

$$t = \frac{x - x_0}{a} \qquad t = \frac{y - y_0}{b} \qquad t = \frac{z - z_0}{c}$$

This creates a three-way equality statement, known as the **symmetric equations** of the line:

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c} \quad \text{where } a, b, c \neq 0$$

If any x , y , or z is 0, then we cannot include the respective expression in our equation. For example, if $a = 0$, our symmetric equations become:

$$x = x_0, \quad \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

This means the line lies completely on $x = x_0$.

Equations of Line Segments

The parametric equations of line *segments* are constrained by the endpoints. Specifically, we can turn a set of parametric equations for a line into a line segment by constraining t to an interval.

For example, the following describes a line segment:

$$x(t) = 5 + t \quad y(t) = 1 + 4t \quad z(t) = 3 - 2t \quad 0 \leq t \leq 1$$

This line segment has endpoints $\vec{r}(0) = (5, 1, 3)$ and $\vec{r}(1) = (6, 5, 1)$.

Derivation: General Equation of a Line Segment

When finding the equations for a line segment given two endpoints and a constraint for t , we must choose the correct direction vector such that the constraints are met.

If given a direction vector $\vec{v}$, we must use $k\vec{v}$ where k is some unknown scale of $\vec{v}$ that will allow our final equations to meet the constraint.

Let a line segment L have endpoints $\vec{r}_0 = \langle x_0, y_0, z_0 \rangle$ and $\vec{r}_n = \langle x_n, y_n, z_n \rangle$. There exists a vector $\vec{v} = \langle a, b, c \rangle$ that is parallel to L . Find a set of parametric equations in t which represent the line segment, where $0 \leq t \leq n$.

$\vec{v} \parallel L \implies k\vec{v} \parallel L \implies \langle ka, kb, kc \rangle \parallel L$. Our vector equation is:

$$\vec{r}(t) = \vec{r}_0 + tk\vec{v} \quad 0 \leq t \leq n.$$

which when written as a parametric is:

$$x(t) = x_0 + akt \quad y(t) = y_0 + bkt \quad z(t) = z_0 + ckt \quad 0 \leq t \leq n.$$

Solve for k when $t = n$ (only one of the below statements are needed):

$$k = \frac{x_n - x_0}{an} = \frac{y_n - y_0}{bn} = \frac{z_n - z_0}{cn}$$

Thus, the equations become:

$$x(t) = x_0 + \frac{x_n - x_0}{n}t \quad y(t) = y_0 + \frac{y_n - y_0}{n}t \quad z(t) = z_0 + \frac{z_n - z_0}{n}t \quad 0 \leq t \leq n.$$

In vector form:

$$\vec{r}(t) = \vec{r}_0 + \frac{\vec{r}_n - \vec{r}_0}{n}t = \vec{r}_0 + \frac{1}{n}\Delta\vec{r}t.$$

Equations of Planes

A plane can also be described using a point which lies on the plane and a direction vector. However, a vector parallel to a plane is not sufficient to describe its direction (there are an infinite amount of planes that would satisfy such condition). Instead, the direction vector is a vector perpendicular to the plane, called the **normal vector** $\vec{n}$.

Derivation: Scalar Equation of a Plane

1. Let the vector $\vec{r}_0 = \langle x_0, y_0, z_0 \rangle$ represent a point that lies on P .
Let the vector $\vec{n} = \langle a, b, c \rangle$ be *orthogonal* to P , indicating its direction.
Let the point represented by $\vec{r} = \langle x, y, z \rangle$ be an arbitrary point which lies on P .
2. The vector pointing from $\vec{r}_0$ to $\vec{r}$, $\vec{r} - \vec{r}_0$, must lie on P because both $\vec{r}_0$ and $\vec{r}$ lie on P .
Since $\vec{n}$ is orthogonal to every vector which completely lies on P , $\vec{n}$ is orthogonal to $\vec{r} - \vec{r}_0$.

The dot product of orthogonal vectors is always 0:

$$\begin{aligned}\vec{n} \cdot (\vec{r} - \vec{r}_0) &= 0 \quad (\text{vector equation}) \\ \langle a, b, c \rangle \cdot \langle x - x_0, y - y_0, z - z_0 \rangle &= 0 \\ a(x - x_0) + b(y - y_0) + c(z - z_0) &= 0.\end{aligned}$$

This is the **scalar equation** of the plane P .

Equation of a Plane

If a plane P in 3-space contains the point represented by position vector $\vec{r}_0 = \langle x_0, y_0, z_0 \rangle$ and the vector $\vec{n} = \langle a, b, c \rangle$ is orthogonal to P , then an arbitrary point represented by position vector $\vec{r} = \langle x, y, z \rangle$ lies on P iff:

$$\begin{aligned}\vec{n} \cdot (\vec{r} - \vec{r}_0) &= 0 \quad (\text{vector equation}) \\ \implies a(x - x_0) + b(y - y_0) + c(z - z_0) &= 0 \quad (\text{scalar equation}).\end{aligned}$$

Example: Equation of a Plane given Point and Normal Vector

The plane P that passes through the point $(2, 4, -1)$. If the vector $2\hat{i} + 3\hat{j} + 4\hat{k}$ is orthogonal to P , find an equation for P .

Using $\vec{r}_0 = \langle 2, 4, -1 \rangle$ and $\vec{n} = \langle 2, 3, 4 \rangle$, we get:

$$2(x - 2) + 3(y - 4) + 4(z + 1) = 0.$$

This can be simplified into standard form ($ax + by + cz = d$):

$$\begin{aligned}2x - 4 + 3y - 12 + 4z + 4 &= 0 \\ 2x + 3y + 4z &= 12.\end{aligned}$$

Example: Equation of a Plane given Three Points

The plane P passes through the following set of points: $\{(1, 3, 2), (3, -1, 6), (5, 2, 0)\}$. Find an equation for P .

Any three points can be used to describe a plane, as long as they are not *colinear*. We can create two vectors out of the three points which are parallel to P :

$$\vec{a} = \langle 3, -1, 6 \rangle - \langle 1, 3, 2 \rangle = \langle 2, -4, 4 \rangle$$

$$\vec{b} = \langle 5, 2, 0 \rangle - \langle 3, -1, 6 \rangle = \langle 2, 3, -6 \rangle$$

The cross product of these two vectors will be orthogonal to P , and we can use it as our normal vector:

$$\begin{aligned}\vec{n} = \vec{a} \times \vec{b} &= \langle 2, -4, 4 \rangle \times \langle 2, 3, -6 \rangle = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -4 & 4 \\ 2 & 3 & -6 \end{vmatrix} \\ &= \begin{vmatrix} -4 & 4 \\ 3 & -6 \end{vmatrix} \hat{i} - \begin{vmatrix} 2 & 4 \\ 2 & -6 \end{vmatrix} \hat{j} + \begin{vmatrix} 2 & -4 \\ 2 & 3 \end{vmatrix} \hat{k} \\ &= 12\hat{i} + 20\hat{j} + 14\hat{k} = \langle 12, 20, 14 \rangle.\end{aligned}$$

Finally, we can select one of the points again to write the equation of P .

Selecting $\vec{r}_0 = \langle 1, 3, 2 \rangle$, we get:

$$12(x - 1) + 20(y - 3) + 14(z - 2) = 0$$

$$12x - 12 + 20y - 60 + 14z - 28 = 0$$

$$12x + 20y + 14z = 100$$

$$6x + 10y + 7z = 50.$$

Intersections of Planes

- The angle of intersection between two planes is the angle between their normal vectors. If the normal vectors of two intersecting planes are $\vec{n}_1$ and $\vec{n}_2$, then the angle θ between them is:

$$\theta = \arccos\left(\frac{\vec{n}_1 \cdot \vec{n}_2}{|\vec{n}_1||\vec{n}_2|}\right).$$

- Two planes can either intersect nowhere, everywhere, or along a single line, called the *line of intersection* between the two planes.
 - If two planes intersect nowhere or everywhere, they are parallel. Two planes are parallel if their normal vectors are also parallel.
- The equations of two intersecting planes create a system of two equations with three unknowns. This means one of our unknowns are free to vary, which is why the set of solutions to the system form a line.
 - The set of parametric equations for the line of intersection can be obtained by parameterizing the free variable.

Example: Line of Intersection of Two Planes

L is the line of intersection between the planes $2x + 3y + 5z = 1$ and $-x + 2y + 3z = 2$. Find a set of parametric equations that represent L .

This is a system of two equations with three unknowns:

$$\begin{aligned}2x + 3y + 5z &= 1 \\ -x + 2y + 3z &= 2\end{aligned}$$

Choose to eliminate two variables. First let's eliminate x :

$$\begin{array}{rcl}2x + 3y + 5z &= & 1 \\ -2x + 4y + 6z &= & 4 \\ \hline 7y + 11z &= & 5\end{array} \implies y = \frac{5}{7} - \frac{11}{7}z.$$

Then, substitute y in terms of z into one of the original equations and solve for x :

$$\begin{aligned}-x + 2y + 3z &= 2 \\ -x + 2\left(\frac{5}{7} - \frac{11}{7}z\right) + 3z &= 2 \\ \frac{10}{7} - \frac{22}{7}z + 3z &= 2 + x \\ -\frac{4}{7} - \frac{1}{7}z &= x.\end{aligned}$$

Now, we have two functions of z : $x(z)$ and $y(z)$. If we let $z = t$, then we have now have a set of three parametric equations for L :

$$x(t) = -\frac{4}{7} - \frac{1}{7}t \quad y(t) = \frac{5}{7} - \frac{11}{7}t \quad z(t) = t.$$

Or, if we let $z = 7t$, we can eliminate a few fractions from our equations:

$$x(t) = -\frac{4}{7} - t \quad y(t) = \frac{5}{7} - 11t \quad z(t) = 7t.$$

Distances Between Planes

The distance between a point and a plane can be calculated using a scalar projection. If P is a point on a plane S , then a vector pointing from Q to P is $\overrightarrow{PQ}$. The shortest distance from Q to S is the scalar component of $\overrightarrow{PQ}$ that is parallel to the vector normal to P .

Distance from a Point to a Plane

Let P and Q be points in $\mathbb{R}^3$, and S is a plane containing P with normal vector $\vec{n}$. The distance from Q to the plane S is given by:

$$|QS| = |\text{comp}_{\vec{n}} \overrightarrow{PQ}| = \frac{|\vec{n} \cdot \overrightarrow{PQ}|}{|\vec{n}|}.$$

Extending this, we can find the distance from a plane A to another plane B ($A \parallel B$) by choosing an arbitrary point on one of the planes and using the above formula.

Skew Lines

Two lines are **skew** if they are not parallel and they do not intersect. Thus, skew lines must lie on parallel but separate *planes*. The distance between the two lines is equal to the distance between the respective parallel planes they lie on.

Example: Verify Two Lines are Skew

Show that the following lines A and B are *skew*:

$$\begin{array}{lll} A : & x = 1 + t & y = -2 + 3t & z = 4 - t \\ B : & x = 2s & y = 3 + s & z = -3 + 4s \end{array}$$

1. First, we can verify that the two lines are not parallel to each other. The direction vectors of A and B are $\langle 1, 3, -1 \rangle$ and $\langle 2, 1, 4 \rangle$ respectively, which are not scalar multiples of each other.
2. Next, we want to show that a point of intersection (x, y, z) does not exist. We can write a system of equations in t and s by equating each component:

$$\begin{aligned} 1 + t &= 2s \\ -2 + 3t &= 3 + s \\ 4 - t &= -3 + 4s. \end{aligned}$$

We can solve for t and s using two of the equations and verify that there is a contradiction when those values of t and s are plugged into the other.

Using the first two equations, we get $t = \frac{11}{5}$ and $s = \frac{8}{5}$:

$$\begin{aligned} 1 + t &= 2s \implies t = 2s - 1 \\ -2 + 3(2s - 1) &= 3 + s \implies s = \frac{8}{5}, t = 2\left(\frac{8}{5}\right) - 1 = \frac{11}{5} \end{aligned}$$

Those values of t and s give us a contradiction in the third equation:

$$\begin{aligned} 4 - \frac{11}{5} &\stackrel{?}{=} -3 + 4\left(\frac{8}{5}\right) \\ \frac{9}{5} &\neq \frac{17}{5} \quad \blacksquare \end{aligned}$$

Since we arrive at a contradiction, A and B do not intersect. Since they are also not parallel, A and B are *skew*.

Distances Between Skew Lines

Derivation: Distance between Two Skew Lines

1. Let lines L_1 and L_2 be skew and lie on planes P_1 and P_2 . P_1 must be parallel to P_2 . Assume L_1 is parallel to $\vec{v}_1$ and L_2 is parallel to $\vec{v}_2$. Finally, let M be a known point on L_1 and thus P_1 .
2. The normal vectors of both P_1 and P_2 must be the same as they are parallel. Since the normal vectors are orthogonal to both $\vec{v}_1$ and $\vec{v}_2$. We can find a normal vector by computing the cross product $\vec{n} = \vec{v}_1 \times \vec{v}_2$.
3. We can now completely define the plane P_1 using point M and normal vector $\vec{n}$. We can now find the distance from an arbitrary point N on L_2 to P_1 :

$$|L_1 L_2| = |\text{comp}_{\vec{v}_1 \times \vec{v}_2} \overrightarrow{MN}| = \frac{|(\vec{v}_1 \times \vec{v}_2) \cdot \overrightarrow{MN}|}{|\vec{v}_1 \times \vec{v}_2|}.$$

Example: Distance between Two Lines

Find the distance between the following two lines, assuming they are skew:

$$\begin{array}{lll} L_1 : & x(t) = 1 + t & y(t) = -2 + 3t & z(t) = 4 - t \\ L_2 : & x(s) = 2s & y(s) = 3 + s & z(s) = -3 + 4s \end{array}$$

The first step is to find the necessary parts needed to define a plane and a point.

Write out the direction vectors, $\vec{v}_1 = \langle 1, 3, -1 \rangle$ and $\vec{v}_2 = \langle 2, 1, 4 \rangle$. The normal vector to both planes containing L_1 and L_2 is:

$$\begin{aligned} \vec{n} &= \vec{v}_1 \times \vec{v}_2 = \langle 1, 3, -1 \rangle \times \langle 2, 1, 4 \rangle \\ &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 3 & -1 \\ 2 & 1 & 4 \end{vmatrix} = \langle 13, -6, -5 \rangle. \end{aligned}$$

We can now define the plane that contains L_1 using the normal vector $\vec{n}$ and a point that lies on L_1 , $\langle 1, -2, 4 \rangle$. Now, find a vector from a point on L_2 to a point on L_1 (and thus the plane containing it):

$$\vec{d} = \langle 0, 3, -3 \rangle - \langle 1, -2, 4 \rangle = \langle -1, 5, -7 \rangle$$

Finally, find the absolute value of the scalar projection of $\vec{d}$ along $\vec{n}$:

$$|L_1 L_2| = |\text{comp}_{\vec{n}} \vec{d}| = \frac{|\vec{n} \cdot \vec{d}|}{|\vec{n}|} = \frac{|\langle 13, -6, -5 \rangle \cdot \langle -1, 5, -7 \rangle|}{\sqrt{13^2 + (-6)^2 + (-5)^2}} = \frac{8}{\sqrt{230}}.$$

Extra: Distance from Point to Line

It can be shown that the distance from a point P to a line L with direction vector $\vec{v}$ is $\frac{|\vec{v} \times \overrightarrow{P_0 P}|}{|\vec{v}|}$.

1.6 Functions and Surfaces

- A scalar function f of n variables assigns each ordered collection of n real numbers $(x_1, x_2, \dots, x_n)$ in a domain $D \subseteq \mathbb{R}^n$ to a single real number $f(x_1, x_2, \dots, x_n)$.
The range of f can be written in set notation as $\{f(x_1, x_2, \dots, x_n) \mid (x_1, x_2, \dots, x_n) \in D\}$.
 - We say a scalar function f of n variables has a mapping $\mathbb{R}^n \mapsto \mathbb{R}$.
- Surfaces in 3-space are represented by functions of two variables, $z = f(x, y)$, where z is a dependent variable and x and y are independent variables.
- We can use **traces** to sketch a graph. By finding the curve of intersection between the function and a set of basic planes ($x = a$, $y = b$, and/or $z = c$), we can sketch the rough shape of a graph.
 - A **horizontal trace** is a trace along a plane $z = c$.
 - A **vertical trace** is a trace along one of the other basic planes.
- A **contour curve** or a **level curve** at k is the specific horizontal trace at $z = k$.
 - A **contour diagram**, where contour curves of varying values of k are overlaid on top of another, can be an alternative way to display a 3D surface in flat contexts.

3D Shapes

- A **cylinder** is defined to be any surface where the cross sectional curve along an axis is constant. In other words, it is a surface defined by at most two of the three variables.
The equation of a cylinder parallel to the xy -plane is $z = f(x)$ or $z = g(y)$.
 - An **elliptical cylinder** is a cylinder where the horizontal cross section is an ellipse.
 - A **parabolic cylinder** is a cylinder where the horizontal cross section is a parabola.
- A **quadric surface** is any surface defined by a second degree linear equation of x , y , or z .
 - In other words, it is a surface defined by an equation which only contains terms that are constant multiples of some second (or lower) degree combination of x , y , and z (ex. x^2 , y^2 , z^2 , xy , xz , yz).
General form of a quadric surface: $ax^2 + by^2 + cz^2 + dxy + exz + fyz + gx + hy + iz + j$ where $a \dots j$ are constants
 - An **ellipsoid** is a quadric surface where all traces are ellipses. A sphere is an example of an ellipsoid. They take the general form $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.
 - An **elliptic paraboloid** is a quadric surface where horizontal traces are ellipses, but vertical traces are parabolas. They take the form $\frac{z}{c} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$.
(Paraboloid opens on z -axis because z is the linear term)
 - A **hyperbolic paraboloid** is a paraboloid where horizontal traces are hyperbolas.
They take the form $\frac{z}{c} = \frac{x^2}{a^2} - \frac{y^2}{b^2}$.
 - A **hyperboloid** is a quadric surface where horizontal traces are ellipses and vertical traces are hyperbolas. Hyperboloids can have either one or two “sheets”.
They take the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = k$, where k can help determine the type of hyperboloid.
The hyperboloid will open along the axis whose term in the above equation is negative.
 - A hyperboloid has only one sheet if $k > 0$, and two sheets if $k < 0$.
 - A hyperboloid is a **cone** if $k = 0$. The sheets meet at a single point.

1.7 Alternative Coordinate Systems

Cylindrical Coordinates

In the **cylindrical coordinate system**, a point in $\mathbb{R}^3$ is represented by a polar coordinate as well as a z -coordinate, represented by (r, θ, z) . The conversion between cylindrical and rectangular coordinates is very similar to how it is done in 2D with polar coordinates:

$$\begin{aligned}(x, y, z) \rightarrow (r, \theta, z) : \quad r^2 &= x^2 + y^2 & \tan \theta &= \frac{y}{x} & z &= z \\(r, \theta, z) \rightarrow (x, y, z) : \quad x &= r \cos \theta & y &= r \sin \theta & z &= z\end{aligned}$$

- Cylindrical coordinates are convenient to describe surfaces that are symmetrical about an axis.
- Cylindrical coordinates need not be oriented via z (e.g. we could have (r, θ, x))

Spherical Coordinates

In the **spherical coordinate system**, a point P in $\mathbb{R}^3$ is described using the radial distance from the point to the origin and two angles, represented by (ρ, θ, φ) , where:

- ρ is the **radial distance** to the origin, which can be computed as $\rho = |\vec{P}|$.
 - It is important to distinguish this from r , the **polar distance**, which is the projection of the radial distance to the xy -plane: $r = \rho \sin \theta$.
- θ is the **polar angle**, the angle used in polar coordinates, computed when P is projected onto the xy -plane.
- φ is the **azimuthal angle**, the angle between the $+z$ -axis and the line connecting P to the origin.

It should also be noted that the restrictions $\rho \geq 0, 0 \leq \theta < 2\pi, 0 \leq \varphi < \pi$ can be used if we want a set of unique points.

Again, the line connecting P to the origin with magnitude ρ makes an angle φ with the $+z$ -axis. We can derive that $r = \rho \sin \varphi$ and $z = \rho \cos \varphi$ are the components of this line. This gives us the conversion into cylindrical coordinates:

$$\begin{aligned}(r, \theta, z) \rightarrow (\rho, \theta, \varphi) : \quad \rho^2 &= r^2 + z^2 & \theta &= \theta & \tan \varphi &= \frac{r}{z} \\(\rho, \theta, \varphi) \rightarrow (r, \theta, z) : \quad r &= \rho \sin \varphi & \theta &= \theta & z &= \rho \cos \varphi\end{aligned}$$

We can then substitute into the conversions between cylindrical and rectangular coordinates to get the conversions between spherical and rectangular coordinates:

$$\begin{aligned}(x, y, z) \rightarrow (\rho, \theta, \varphi) : \quad \rho^2 &= x^2 + y^2 + z^2 & \tan \theta &= \frac{y}{x} & \tan \varphi &= \frac{\sqrt{x^2 + y^2}}{z} \\(\rho, \theta, \varphi) \rightarrow (x, y, z) : \quad x &= \rho \sin \varphi \cos \theta & y &= \rho \sin \varphi \sin \theta & z &= \rho \cos \varphi\end{aligned}$$